Exp 13

29/9/26

Internetworking with routers in CISCO PACKET TRACER Simulator

A) Internetworking Using Router

Aim:- To design and configure a simple internetwork using a router in Cisco Packet Tracer and verify connectivity between two PCs.

Observation:-

1) A router and two PCs are used to create a simple internetwork.

2) PC0 is connected to the 1st interface of Router 1 using a copper straight-through cable

3) PC1 is connected to the second interface of Router 1.

4) Router interfaces are assigned different network addresses.

- 192.168.10.1/24
- 192.168.20.1/24

5) PC0 is assigned IP address 192.168.10.2 with gateway 192.168.10.1

6) PC1 is assigned IP address 192.168.20.2 with gateway 192.168.20.1

7) The router forwards packets between two networks

8) A PDU is sent from PC0 to PC1 to verify network connectivity

Result:- Thus, a simple internetwork was successfully designed and configured using a router and communication between PC0 and PC1 was successfully verified.

B) Wireless Router, DHCP Server and Internet Cloud

Aim: To design and configure an internetwork using a wireless router, DHCP server and Internetwork cloud.

Observations :-

1) A wireless router, PC, Laptop, cable modem, Internet cloud and cisco.com server are connected
2) The wireless router is configured with the SSID Home network
3) DHCP is enabled on the wireless router.
4) The PC obtains its IPv4 address automatically through DHCP
5) The laptop is configured with a wireless module and connected to homenetwork
6) The internet cloud is connected to the cable modem.
7) The Cisco.com server is configured as a DHCP and DNS server.
8) The PC receives an address in the 192.168.0.* range.

9) Connectivity is verified by using ping cisco.com.

Result:- Thus, the wireless routed DHCP server and internetwork cloud were successfully configured and connectivity to cisco.com server was verified.